

Project brief: a complaint-intelligence assistant

To · Data Engineering

From · Analytics Team

Re · A prototype we'd like you to build

Time · One week — a few focused days of work

The short version. We're a consumer-finance analytics company. Our analysts spend their days answering questions about consumer complaints — for our clients, for our own research, for the occasional regulator. Today they do it by hand: filtering spreadsheets, skimming thousands of complaint narratives, eyeballing trends and writing them up. It's slow, it doesn't scale, and two analysts asked the same question can come back with different answers. We want an assistant they can just **ask** — plain English in, a grounded answer out. This brief tells you what our analysts need and what we'd call a good result. *How* you build it is your call.

First, the data — go and get it. We work with the **CFPB Consumer Complaint Database**, the US consumer-finance regulator's public archive of real complaints. We are deliberately not handing you a file: finding the public source, choosing how to pull from it (there's more than one way), and building a **pipeline from the source into our environment** is the first part of the job — it's the part we do every week. Scope your extract to complaints **received 2024-01-01 through 2025-12-31 that carry a consumer narrative** (the consumer's own written account — that's the text our analysts live in; expect the extract to run to a couple of million records). Pull once, snapshot it, and build everything on that snapshot — your numbers need to be reproducible, not chasing a database that updates daily.

Organize it the way our lakehouse is organized. Our data estate follows the **medallion convention**, and your pipeline should too, with the layers laid out as folders — plain folders on your own machine are fine, we're not asking for real cloud storage here:

- **bronze/** — the raw data exactly as fetched, untouched, plus a small manifest of what you pulled, from where, when, and how much of it there is. If we ever doubt a number, this is where the audit trail starts.
- **silver/** — cleaned and conformed: proper types, parsed dates, normalized company names, duplicates dealt with, one trustworthy record per complaint.
- **gold/** — analysis-ready: the tables the assistant actually queries (think complaint volumes and response-timeliness by company, product, and month — the shapes our analysts' questions take).

Each layer should be rebuildable from the one before it. Keep the plumbing in proportion — it must be sound and auditable, but the assistant is the deliverable.

What our analysts actually ask. Two flavours, and they blur together constantly:

- *"What are people saying about..."* — the themes, the anger, the specific failures customers describe. *What's going wrong for people with debt collectors? What do borrowers hate about their student-loan servicer?*
- *"How many / which / how much..."* — counts, rates, rankings, trends. *Which company's complaints jumped the most last quarter? Who's slowest to respond?*

And, most usefully, **both at once**: *"Which company had the biggest spike in complaints in Q3, and what were people actually complaining about?"* — a number **and** the story behind it, in one breath. Their questions also carry hard constraints — a specific company, product, state, or date window — sitting on top of a fuzzy topic. *"Zelle scam complaints from Texas this year"* is a completely normal ask, and both halves have to work together.

Two engines under one hood. We expect the "how many / which / how much" questions to be answered by **querying your gold tables directly** — not guessed from a handful of narratives. An analyst asks in plain English; the tool turns that into a real query, runs it, and — this part matters — **shows the query it ran**. When a figure lands in a client report, someone always asks how we got it, and *"the model said so"* isn't an answer we can give. So think of it as one tool with two ways of reaching the data: **reading the narratives** for the *"what are people saying"* questions, and **querying the numbers** for the *"how many / which"* ones. The tough questions lean on both at once, so the two need to talk to each other.

What "good" looks like to us.

- **It's honest.** Analysts put these answers in front of clients and executives. A confident wrong answer is worse than *"I don't know."* When the answer isn't in the data, the tool must say so plainly. (*Heads-up: a few of the 15 questions below are ones we know this data can't answer. We put them there on purpose — we want to see what the tool does when it's got nothing to stand on. We won't tell you which ones.*)
- **It shows its work.** Every claim points back to its evidence — the specific complaints by their Complaint ID for anything read from the narratives, and the query itself for anything counted, ranked, or aggregated. An analyst should be able to open the originals or re-run the number. *"Trust me"* doesn't fly here.
- **It survives real data.** This archive is real and really messy: companies appear under **legal names** while customers ask about brands (*"Cash App"* is in there as *"Block, Inc."*, *"Zelle"* as *"Early Warning Services, LLC"*); narratives carry **XXXX redactions**; bulk-filed near-duplicate form letters abound; the usual grime of blank fields, inconsistent casing, and dates stored as text is everywhere. Handling this well is what silver is for.
- **You can tell us it works.** We don't just want something that runs — we want to know *how well* it works and how you convinced yourself of that.

How we'll judge it. Attached is `questions.json` — 15 real questions in the style our analysts ask. We'd like your assistant's answers to all fifteen (in the simple format the Reference Pack describes — generated by your system, not hand-written), plus whatever measurement you did to check its quality. Then we'll sit down for **30–45 minutes** and have you walk us through it live: the pipeline, the design, a few queries we'll throw at you on the spot, your results, and the thinking behind your choices. Bring it runnable — we'll ask for a small change or two on the day. That session matters as much as the code; it's where we understand how you think.

Practicalities.

- **Where to spend your energy:** the assistant is the deliverable; the medallion pipeline should be sound and auditable without being elaborate.
- **Scale sensibly.** Nobody expects you to embed two million narratives on a take-home budget. Choose a **documented working set** for semantic search (or a retrieval strategy that narrows before it embeds) — just make the choice deliberate, reproducible, and written down. One warning: the 15 questions probe specific companies, products, states, and months; a working set that silently drops those slices will fail questions your pipeline actually contains the answers to.
- **Time:** you have a week, but this shouldn't need more than a few focused days. We don't reward heroics, and we'd rather see good judgement about what to cut than an exhausted kitchen-sink submission. If you drop something, tell us what and why.
- **Tools:** anything you like — any language, storage, orchestration, or models, hosted or local. Use AI coding assistants freely; just be ready to explain and defend every decision at the demo.
- **A first honest cut beats a clever fragile one.** We care more that this is traceable and truthful than that it's sophisticated.

Questions about any of this are welcome — just reach out. We're looking forward to seeing what you build.

— The Analytics Team, Meridian Insights